

ASOSAN, Japan

WMO: 478210

Lat: 32.8804N

Lon: 131.0733E

Elev: 1144

StdP: 88.31

Time Zone: 9.00 (E09)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.0	-7.7	-18.9	0.8	2.9	-15.1	1.2	2.3	12.4	0.0	11.2	-0.7	6.3	320	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	25.8	21.0	24.7	20.5	23.6	20.0	21.9	24.2	21.3	23.3	20.8	22.5	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.3	22.8	20.7	17.7	22.1	20.3	17.2	21.6	70.3	24.3	68.1	23.4	66.2	22.7	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.2	10.5	9.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-11.2	27.4	1.6	0.9	-12.3	28.0	-13.3	28.6	-14.2	29.1	-15.3	29.8		
			-11.5	23.0	1.6	0.7	-12.7	23.5	-13.6	23.9	-14.5	24.3	-15.7	24.8		
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-1.4	0.3	3.7	8.9	13.9	17.0	20.6	21.2	18.2	12.9	7.0	1.0
	DBStd	8.47	3.91	4.42	4.14	3.65	2.49	1.92	1.72	1.47	2.54	3.08	3.99	4.15
	HDD10.0	1283	354	272	198	61	3	0	0	0	11	106	279	
	HDD18.3	3125	613	506	454	282	139	48	3	1	33	169	339	538
	CDD10.0	1401	0	1	2	28	123	209	328	348	246	99	16	0
	CDD18.3	201	0	0	0	0	1	8	72	91	29	1	0	0
	CDH23.3	290	0	0	0	0	2	3	122	143	20	0	0	0
	CDH26.7	8	0	0	0	0	0	0	4	4	0	0	0	0
Wind	WSAvg	4.5	4.8	4.8	5.0	4.9	4.5	4.6	4.7	4.3	4.3	4.1	4.1	4.6
Precipitation	PrecAvg	2750	81	115	179	203	242	588	502	281	269	126	107	75
	PrecMax	4853	167	227	472	372	434	1264	1115	726	718	298	244	149
	PrecMin	1565	3	37	73	73	82	186	110	64	48	10	22	4
	PrecStd	617	40	51	81	81	98	259	285	158	172	76	56	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	12.9	14.9	19.7	23.2	23.7	27.3	27.1	25.3	21.3	16.8	11.9
		MCWB	8.8	10.0	8.9	12.6	13.6	17.9	21.2	21.6	20.2	16.1	12.9	10.1
	2%	DB	8.3	10.9	12.9	17.2	21.0	22.2	25.9	25.9	23.7	19.4	15.3	10.0
		MCWB	6.5	9.2	9.0	11.1	13.1	16.9	21.0	21.2	19.4	15.2	12.7	7.7
	5%	DB	6.4	9.2	11.3	15.7	19.4	20.9	24.7	25.0	22.6	18.2	14.1	8.5
		MCWB	5.0	7.6	8.3	11.3	12.9	17.0	20.7	20.9	18.8	15.1	11.9	6.5
	10%	DB	4.4	7.1	9.9	14.3	18.1	19.9	23.5	23.9	21.4	17.2	12.7	7.2
		MCWB	2.5	5.2	7.5	10.4	13.2	17.6	20.3	20.5	18.5	14.6	10.7	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	11.7	12.7	16.2	18.2	20.7	22.7	22.7	21.6	18.6	15.6	11.0
		MCDB	10.2	11.9	13.0	16.9	18.6	21.6	25.3	25.5	23.8	19.6	15.8	11.2
	2%	WB	7.5	9.9	11.0	14.7	17.2	19.8	21.8	22.1	20.6	17.5	14.0	8.9
		MCDB	7.8	10.2	11.5	15.4	17.8	20.3	24.2	24.5	22.1	18.5	14.5	9.2
	5%	WB	5.3	8.1	9.6	13.2	16.1	19.3	21.3	21.5	20.0	16.6	12.6	7.3
		MCDB	5.9	9.1	10.4	14.1	17.1	19.7	23.2	23.6	21.3	17.5	13.5	8.0
	10%	WB	3.0	5.9	8.1	11.7	15.0	18.7	20.8	21.0	19.5	15.4	11.3	5.6
		MCDB	3.9	6.8	9.1	13.1	16.3	19.2	22.4	22.9	20.6	16.4	12.3	6.7
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.8	7.5	7.5	7.2	5.2	4.8	5.2	5.6	6.4	6.3	5.9
		MCDBR	7.4	8.4	8.5	8.9	9.1	7.0	7.0	6.8	6.5	6.7	6.7	7.2
		MCWBR	7.3	7.6	7.0	6.7	6.2	4.7	4.3	4.1	4.1	5.1	5.6	6.7
	5% WB	MCDBR	7.0	7.6	7.5	6.5	6.0	4.1	5.9	5.9	5.1	5.1	5.6	6.5
		MCWBR	7.6	7.8	7.6	5.9	5.6	3.4	3.8	3.7	3.5	4.1	5.4	6.8
Clear-Sky Solar Irradiance	taub	0.342	0.373	0.419	0.435	0.443	0.453	0.458	0.440	0.414	0.376	0.348	0.336	
	taud	2.304	2.242	2.093	2.104	2.142	2.196	2.232	2.281	2.319	2.350	2.370	2.341	
	Ebn at Noon	863	871	854	857	850	838	831	842	850	858	852	848	
	Edh at Noon	104	123	154	159	155	146	140	132	122	109	97	95	
All-Sky Solar Radiation	RadAvg	2.46	3.08	4.09	4.80	5.08	4.17	4.87	5.03	4.13	3.51	2.69	2.25	
	RadStd	0.23	0.28	0.26	0.47	0.58	0.62	0.66	0.48	0.46	0.38	0.27	0.16	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page